Rep Note

Jiawen Huang

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1 Assumption

We are working on $\mathbb{C}$ and finite group G.

2 Irreducible and Decomposable Representation

Counterexample:

$$\phi : \mathbb{C} \to GL_2(\mathbb{C})$$

$$\phi(a) = \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix}$$

This is irreducible but not decomposable.

2.1 Unitary Representation

Definition 2.1. Given a rep $\phi : G \to GL(V)$ and V is an inner product space. ϕ is irreducible if

$$\langle \phi_g(v), \phi_g(w) \rangle = \langle v, w \rangle$$

In $\mathbb{C}$, the standard inner product is defined as $\langle v, w \rangle = \sum \overline{v_x} w_x$

Proposition 2.2

If rep ϕ is unitary, then ϕ irreducible if and only if ϕ is indecomposable.

Proof. If ϕ is reducible, then let W be the G invariant subspace of V, that is $\phi_g(W) = W$ for all g. Consider the subspace $W^\perp$. Notice that for any $x \in W^\perp$, any $v \in W$ and any $g \in G$

$$0 = \langle x, \phi_{g^{-1}}(v) \rangle = \langle \phi_g(x), \phi_g \phi_{g^{-1}}(v) \rangle = \langle \phi_g(x), v \rangle$$

Thus, for any $x \in W^\perp$, $\phi_g(x)$ is orthogonal to v for all $v \in W$ and $g \in G$. Thus, $W^\perp$ is a G invariant subspace of V. Thus, $V = W \oplus W^\perp$ as rep. $\square$

Remark 2.3. We would like these two conecpts to be equivalent. So we want our rep to be unitary, or equivalent to a unitary rep.

Theorem 2.4

Every representation of a finite group G is equivalent to a unitary rep. From now on, we assume all the reps are unitary.

Proof. Let $\rho : G \to GL(V)$ be a rep. It is similar to $\phi : G \to GL(\mathbb{C}^n)$ where $\mathbb{C}^n$ has standard basis. We can find an inner product of $\mathbb{C}^n$. We define a new inner product of the space.

$$(v, w) = \sum_{g \in G} < \phi_g(v), \phi_g(w) >$$

This is well defined because G is finite. And we can verify (v, w) is indeed an inner product of $\mathbb{C}$.

It's easy to see ϕ is a unitary rep w.r.t. the inner product (v, w). □

Theorem 2.5 (Maschke)

Every rep of a finite group can be decomposed into irreducible rep.

Proof. By theorem 2.4 and proposition 2.2, rep of a finite group is irreducible is equivalent of being indecomposable. □

3 Morphism of Reps

Does any linear transformation is morphism of rep? No, but we can turn it into a morphism of rep.

Definition 3.1. Given rep $\phi : G \to GL(V)$ and $\rho : G \to GL(W)$. For any linear transformation T from V to W, we define

$$T^{\#} := \frac{1}{|G|} \sum_{g \in G} \rho_{g^{-1}} T \phi_g$$

Proposition 3.2

- If T is a morphism of reps, then $T = T^{\#}$

Lemma 3.3 (Schur's Lemma)
The morphism between two irreducible reps ϕ and ρ is either 0 or invertable. Consequently:

- If $\phi \not\sim \rho$, then $\text{Hom}_G(\phi, \rho) = 0$
- If $\phi = \rho$, then for any $T \in \text{Hom}_G(\phi, \rho)$, $T = \lambda I$
- if $\phi = \rho$, for any linear trasnformation L, $L^\# = \frac{Tr(L)}{deg(\phi)} I$

Corollary 3.4
Let G be an abelian group, irreducible rep of G must be degree of one.

For any $h \in G$, ϕ_h commutes with ϕ_g for any $g \in G$, then $\phi_h \in \text{Hom}_G(\phi, \phi)$. Then use schur lemma.

Corollary 3.5
Any matrix with finite order must be diagonalizable.

4 The Orthogonality Relations

Definition 4.1. Given a finite group G, define

$$L(G) = \mathbb{C}^G = \{f \mid f : G \to G\}$$

This is in fact a $\mathbb{C}$ vector space and we can define the inner product as

$$< f_1, f_2 >= \frac{1}{|G|} \sum_{g \in G} f_1(g) \overline{f_2(g)}$$

Its dimension over $\mathbb{C}$ is the size of G.

Definition 4.2. Given a rep ϕ. Define $\phi_{ij}(g)$ to be the (i,j) entry of the matrix $\phi(g)$. Notice that $\phi_{ij} \in L(G)$.

Theorem 4.3 (Schur orthogonlaity relations)
Suppose ϕ and ρ are inequivalent irreducible unitary representation. Then,

- $< \phi_{ij}, \rho_{k\ell} >= 0$
- $< \phi_{ij}, \phi_{k\ell} >= \begin{cases} \frac{1}{n} & \text{if } i = k, j = \ell \\ 0 & \text{else} \end{cases}$

Proof. The key fact to prove this is $(E^{\#}_{ik})_{jl} =< \phi_{ij}, \rho_{kl} >$. We then use schur lemma. $\square$

Thus, we know that ϕ_{ij} form an orthonormal set of $L(G)$. Thsi implies

$$\sum_{\text{irreducible } \phi} \deg(\phi)^2 \leq |G|$$

With this, we are ready to develop character theory.

5 Character Theory

Definition 5.1.

$$\chi_\phi(g) = Tr(\phi(g)) = \sum_{i=1}^{n} \phi_{ii}(g)$$

Notice that χ_ϕ is in $L(G)$.

Proposition 5.2

All of them are trivial by the definition

- $\chi_\phi(1) = \deg(\phi)$
- If $\phi \sim \rho$, $\chi_\phi = \chi_\rho$
- $\chi_\phi(gxg^{-1}) = \chi_\phi(x)$, χ_ϕ is a class function.
- $\chi_{\phi\oplus\rho} = \chi_\phi + \chi_\rho$

Denote $Z(L(G))$ as the set of all class functions

Proposition 5.3

$\dim(Z(L(G))) = \#$conjuagcy classes of G

Trivial.

Theorem 5.4

$< \chi_\phi, \chi_\rho >= 0$ iff $\phi \not\sim \rho$
$< \chi_\phi, \chi_\rho >= 1$ iff $\phi \sim \rho$

Consequently:

1. Character fully characterizes the equivalent classes of reps
2. Notice that χ_ϕ is a class function so the number of irreducible reps $\leq$ number of conjugacy classes of G.

Proof is easy by Theorem 4.3.

We have shown that a finite group rep has a decomposition. Is it unique? YES.

Theorem 5.5

Every finite group representation can be uniquely decomposed into irreducible representations.

Corollary 5.6

A rep ϕ is irreducible if and only if $< \chi_\phi, \chi_\phi >= 1$

This gives us a way to decide whether a rep is irreducible or not.

Problem 5.7. Find all the irreducible reps of S_3.

1. $\sum \chi_\phi(1)^2 = \sum \deg(\phi)^2 \leq 6$
2. Trivial rep

This is tough!

6 Regular Rep L

This is the rep of G on the group algebra $\mathbb{C}[G]$. This is nice because we can compute $\chi_L(g) = \begin{cases} |G| & \text{if } g = 1 \\ 0 & \text{else} \end{cases}$

Thus, $< \chi_L, \chi_{\phi_i} >= m_i = d_i$

Corollary 6.1

$$|G| = \sum \deg(\phi)^2$$

And $\phi_{ij}^{(k)}$ form a basis of $L(G)$.

Furthermore, we can show that χ of all irreducible reps form a basis of $Z(L(G))$ (This is a fairly complicated result-Artin Theorem). This concludes that the number of conjugacy classes = the number of irreducible reps.